What is Terraform?

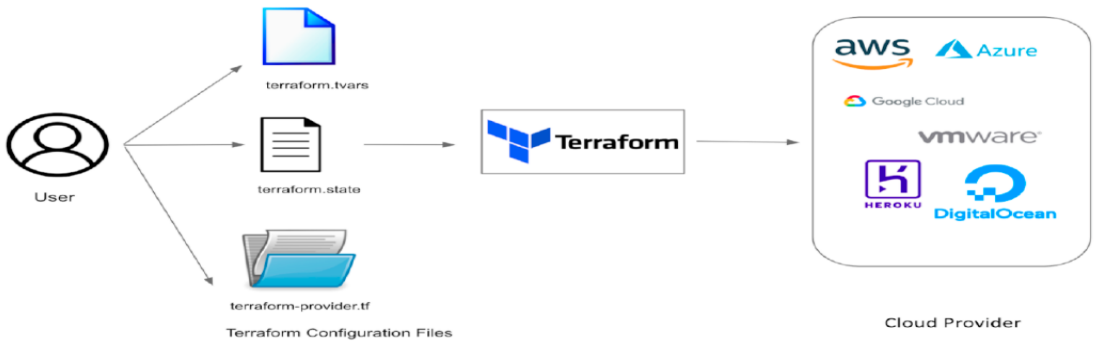
- Terraform is an **infrastructure as a code** tool that lets you build, change and version control cloud resources safely, easily and efficiently
- Terraform is an open source "Infrastructure as a Code" tool, created by HashiCorp.
- Terraform is developed using Go Language in the year 2014.
- All the configuration files used HCL (HashiCorp Configuration Language) language for the code.
- Terraform uses a simple syntax, can provision infrastructure across multiple clouds & On premises.
- Terraform is cloud-agnostic and supports multiple cloud providers such as AWS, Azure, Google Cloud, and others

Infrastructure as Code(IaC)

Before the invention of IaC, infrastructure management was typically a manual and time-consuming process.

System administrators and operations teams had to:

1. Manually Configure Servers
2. Lack of Version Control
3. Documentation Heavy
4. Limited Automation
5. Slow Provisioning



Alternatives Of Terraform:

CLOUD	IAC tool (Native)	Configuration written in
AWS	CFT (cloud formation template)	JSON / YAML
AZURE	ARM TEMPLATES	JSON
GCP	CLOUD DEPLOYMENT MANAGER	YAML / PYTHON

Features of Terraform:

Terraform is a tool that helps you set up and manage cloud infrastructure using code.
It lets you define how the infrastructure should be, and Terraform will take care of setting it up.

Multi-Cloud Support:

- Terraform can works with different cloud platforms like AWS, Azure, Google Cloud, and others.
- It can also be used with on-premise servers or a mix of both.

Declarative Configuration:

In simple terms, you write down how you want your infrastructure to look in a configuration file (using a special language called HCL). and Terraform takes care of figuring out how to achieve that state.

Resource Provisioning:

- Terraform helps you create and manage things like virtual machines, storage, and networks and more.
- It creates and updates resources based on the configuration provided.

State Management:

- Terraform maintains a state file that keeps track of the current state of the infrastructure.
- This file is used to plan and apply changes, ensuring that Terraform can update resources accurately.

Plan and Apply Workflow:

- Before changing anything, Terraform creates a plan showing what it will do.
- You can review this plan and then apply it to make the changes.

Version Control Integration:

- You can store your Terraform configuration files (except state files) in version control tools like Git.
- This helps you work together with others, check for changes, and keep track of updates.

Modular Configuration:

- You can organize your infrastructure setup into modules.
- modules make easier to reuse and share configuration files across different projects.

Community support:

Terraform has a big community and many ready-made solutions for common infrastructure needs, which makes it easier for you to get started.

TERRAFORM LIFECYCLE:

The Terraform lifecycle refers to the sequence of steps and processes that occur when working with Terraform to manage infrastructure as code.

Terraform Lifecycle

Write your infrastructure in Configuration file.

Users define their infrastructure in a declarative configuration language, commonly using HashiCorp Configuration Language (HCL).

Initialize (terraform init):

terraform init command will initialize a Terraform working directory.

This command initializes the Terraform working directory & also downloads provider plugins.

Plan (terraform plan):

terraform plan command creates an execution plan.

Terraform compares the desired state from the configuration with the current state and generates a plan for the changes required to reach the desired state.

Review Plan: Examine the output of the plan to understand what changes Terraform intends to make to the infrastructure. This is an opportunity to verify the planned changes before applying them.

Apply (terraform apply):

Execute terraform apply command to applies the changes outlined in the plan.

Terraform makes the necessary API calls to create, update, or delete resources to align the infrastructure with the desired state

Destroy(terraform destroy) :

When infrastructure is no longer needed, or for testing purposes, run terraform destroy to tear down all resources created by Terraform. This is irreversible, so use with caution.

Create an AWS IAM User:

To interact with AWS programmatically, you should create an IAM (Identity and Access Management) user with appropriate permissions.

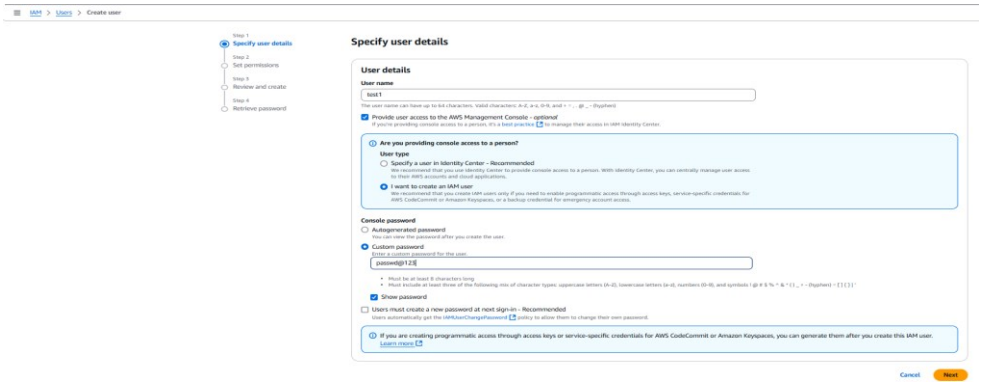
Steps:

a. Log in to the AWS Management Console with an account that has administrative privileges.

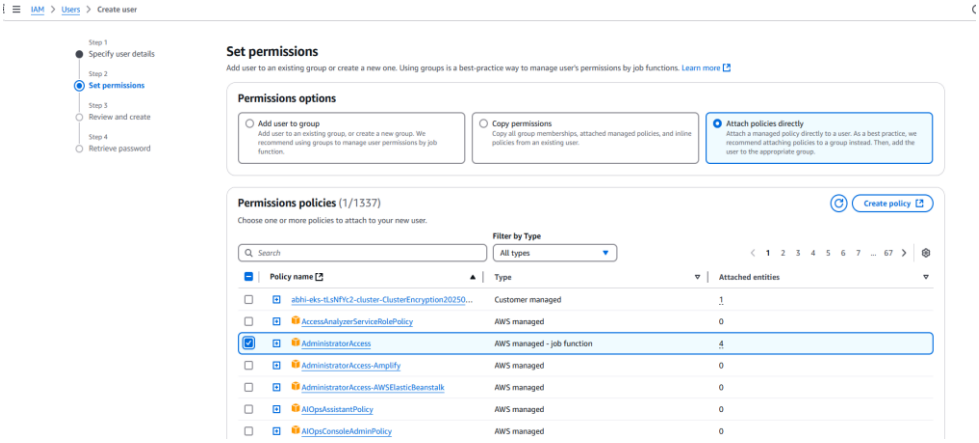
b. Navigate to the IAM service.

c. Click on “Users” in the left navigation pane and then click “Add user.”

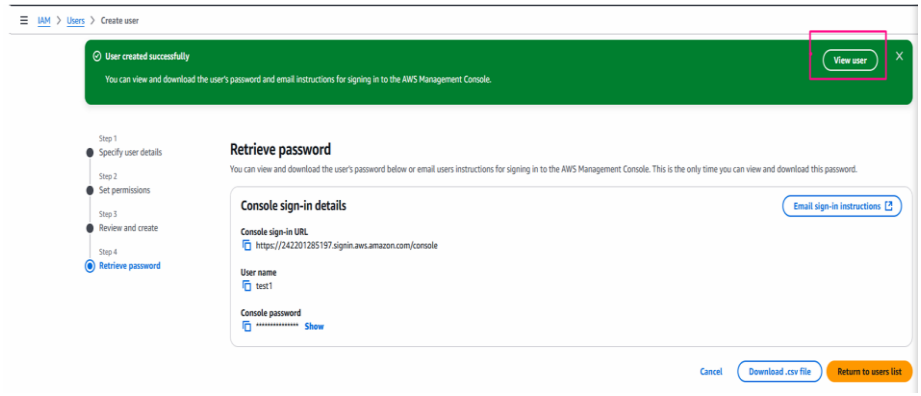
Give username like **tf-user-1**



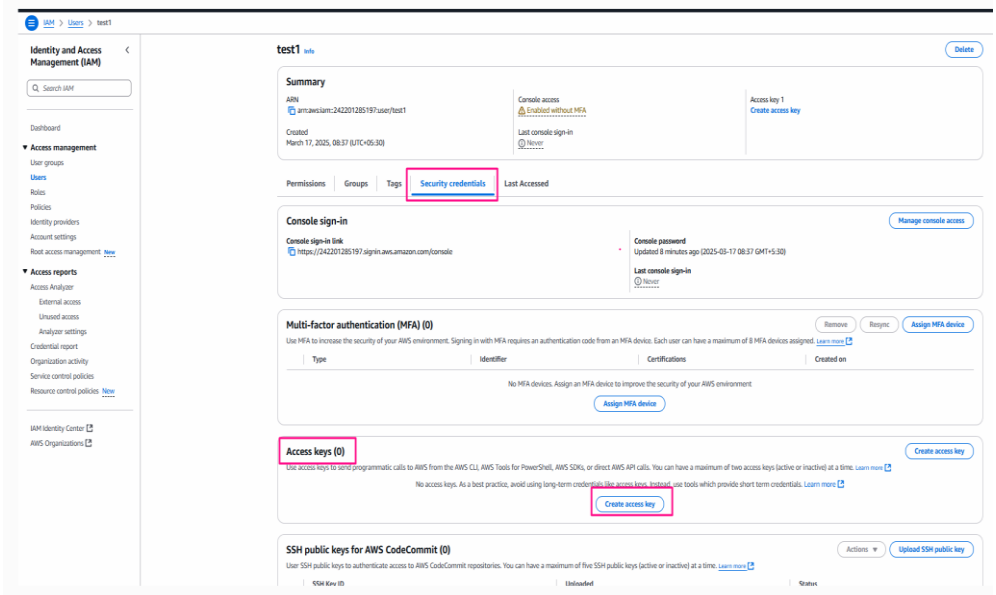
Attach policy to user – add **AdministratorAccess** policy

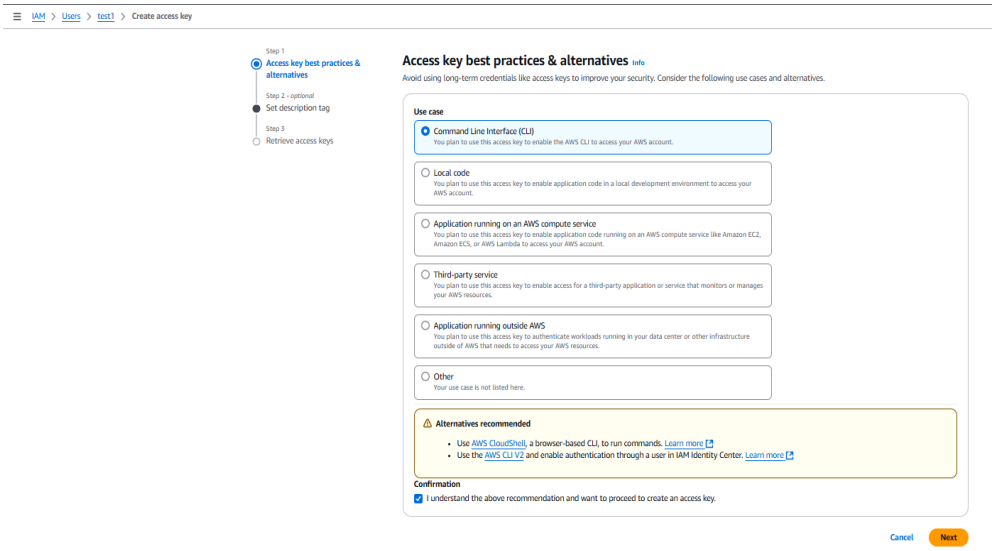


Click on view user

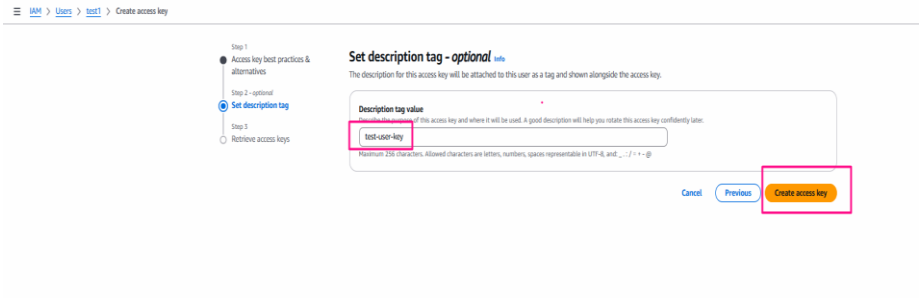


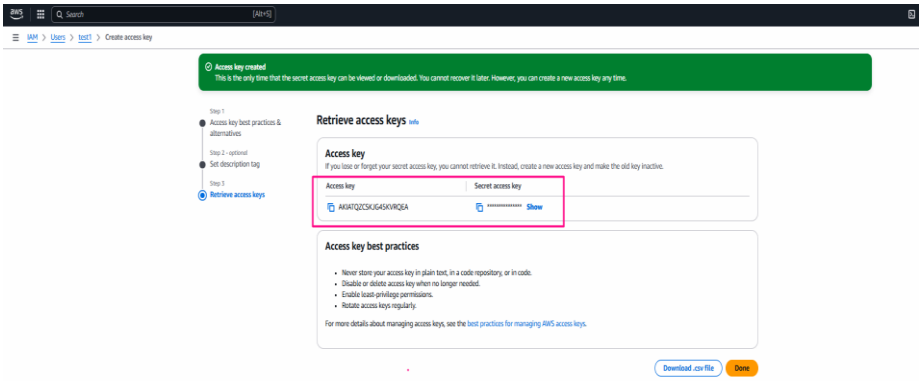
Create an access keys for user





Add tag for reference





Make sure to **save the Access Key ID and Secret Access Key** that are displayed after user creation;

Install Terraform:

Installation steps - https://developer.hashicorp.com/terraform/install

Launch ec2instance (t2.micro) with amazonlinux flavour & install terraform from below steps

```
sudo yum install -y yum-utils shadow-utils
sudo yum-config-manager --add-repo https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
sudo yum -y install terraform
```

verify installation:

```
[ec2-user@ip-172-31-26-175 ~]$ terraform -version
Terraform v1.11.2
on linux_amd64
[ec2-user@ip-172-31-26-175 ~]$
```

Note: Install **terraform** plugin in visual studio

Ex1. Write an terraform file to Create an VM in aws cloud (ec2 instance)

Create a directory called ec2instance & inside that create a .tf file

```
provider "aws" {
    # in this block we will specify which cloud we are using, which region &
credentails
    region = "us-east-1"
    access_key = "AKIAUEBQGJSOXDQ6GPFG"
    secret_key = "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"
}

resource "aws_instance" "my-instance-1" {
    ami = "ami-04b4f1a9cf54c11d0"
    instance_type = "t2.micro"
    key_name = "terraform-kp-1"

}
```

Practical's & observation:

Step1: after creating file in directory run ***terraform init*** to initialize terraform & terraform also download AWS plugin file

Step2: run ***terraform validate*** command, it will check if the syntax is valid or any changes required.

Step3: run ***terraform plan*** command, check for the output for plan

Step4: run ***terraform apply*** command, which will create the resource (ec2 instance) specified in the .tf files

Step5: run ***terraform destroy*** command, which will delete all the resources created from that .tf file

Explanation of above Terraform script is used to provision an AWS EC2 instance.

Provider Block:

```
provider "aws" {
    region = "us-east-1"             # Specifies the AWS region where resources will be created.
    access_key = "AKIAUEBQGJSOXDQ6GTMG" # AWS access key for authentication (this key is specific to your AWS account).
    secret_key = "a/v7k9ZAsZasQKH3WNo4s95/0sB5ZCNrlOTi/Kj1"  # AWS secret key for authentication (this key is specific
to your AWS account).
}
```

- **Provider**: This tells Terraform which cloud platform you're using. In this case, it's AWS (Amazon Web Services).
- **Region**: It specifies that the resources should be created in the us-east-1 region of AWS.
- **Access Key** and **Secret Key**: These are the credentials used by Terraform to authenticate and interact with AWS. In practice, **you should avoid hardcoding keys** in the script for security reasons.

Resource Block:

```
resource "aws_instance" "my-instance-1" {
    ami = "ami-04b4f1a9cf54c11d0"  # The Amazon Machine Image (AMI) ID used to create the instance.
    instance_type = "t2.micro"      # Specifies the type of instance (in this case, a small instance).
    key_name = "terraform-kp-1"     # The name of the key pair to access the instance once it's running.
}
```

- **Resource**: This block defines a resource that will be created. Here, it's an **EC2 instance** (aws_instance).
- **AMI**: The AMI ID ami-04b4f1a9cf54c11d0 specifies which operating system and software setup will be used for the instance.

- **Instance Type**: t2.micro is a small & free instance type on AWS.
- **Key Name**: The key pair (terraform-kp-1) is used to securely access the EC2 instance via SSH after it is created.

In summary:

- This Terraform script is setting up an EC2 instance in AWS using a specific AMI (operating system image), instance type (t2.micro), and key pair (terraform-kp-1) for SSH access.
- The provider block tells Terraform to interact with AWS and specifies the credentials and region to be used.

Important Note:

For security reasons, it's highly recommended **not to hardcode your access and secret keys** directly in your Terraform script. Instead, use environment variables, AWS IAM roles, or other secure methods to provide credentials.